

Clock and reset in the host

Every Verilator run needs a disciplined clock and reset story

Assert, wait, release

- Assert reset low or high consistently with your active level
- Count posedges while reset is held, often two or three is enough for literacy
- Release reset synchronously relative to the clock edge you care about
- After release, let the DUT run

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / TB clock + reset patterns

INTERACTIVE TOOL

TB clock + reset patterns

Sketch TB **clock generation** and **reset assert / deassert** on a short timeline — how many cycles of `rst_n=0`, then release relative to posedge `clk`. Starter: hold reset two cycles, then release.

Starter example: free-running `clk`; `rst_n` low for two posedges, then release (classic sync-style).

Load starter example

Challenges 0 / 22 cleared

Quiz: **clock**: A typical TB clock is generated with...

☐ a forever toggle (or equivalent) in the testbench

☐ only the DUT's clock-gating cell

☐ \$finish inside the clock block

☐ randomize() on clk each ns

Show hint

Check

Next

Idle

1

2

3

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5

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22

Core ideas

forever clk

TB clock is a free-running toggle, not a DUT output.

Assert

Drive `rst_n=0` for N clock cycles at bring-up.

Sync release

Deassert near a clock edge so flops see a clean sample.

Async release

Change reset between edges — still legal; know the hazard.

Lab

Lab starter

learn_verilator — tb clock reset

Real Verilator practice

- In Track A, open EXAMPLES and implement or inspect a host loop that toggles clock with
- Run once and confirm outputs change only after reset deasserts
- Say how many posedges you waited, make the number intentional

Pitfalls to watch

- Do not hold reset forever and call it a passing test
- Do not release reset asynchronously if your DUT expects synchronous deassert
- Do not forget the clock entirely in a C++ host, eval needs input changes
- And do not copy a browser sketch verbatim without matching your DUT's reset polarity

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, reach ready with a synchronous release pattern
- In Track A, run one reset sequence you can explain out loud
- When you are ready, take the short quiz, then continue to Verilator trace